

# SahaPilot: A Low-Cost ESP32-Based Autonomous Ground Vehicle with RF-Based Intrusion Detection for Small-Scale Industrial Logistics

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## Abstract

*Commercial automated guided vehicles (AGVs) remain financially out of reach for most small and mid-sized manufacturers, and rarely include any physical-security layer for the environment they operate in. This report describes SahaPilot, a low-cost autonomous ground vehicle built on ESP32 hardware that combines line-following navigation with obstacle avoidance and an RF-based intrusion detection layer that flags unauthorized wireless devices by their MAC-address OUI (Organizationally Unique Identifier). Before committing to hardware, the control strategy and detection logic were validated in software: a proportional-derivative (PD) controller is shown, via simulation, to converge roughly 7-8x faster than bang-bang or proportional-only control; a consecutive-scan confirmation rule for the RF layer is shown to eliminate false positives while adding only a 2-3 scan detection delay. A navigation logic defect was also identified through randomized testing -- not code inspection alone -- and corrected, reducing simulated collisions from 1,009 to 0 across 200 trials. This report documents the methodology, results, and the project's transition from simulation to a physical prototype.*

## 1. Introduction

Industrial AGV/AMR systems typically cost between 15,000 and 40,000 USD per unit and require dedicated infrastructure, placing them out of reach for small and mid-sized manufacturers (KOBİ in the Turkish context). Simpler line-following robots exist at far lower cost, but they offer no protection against unauthorized human or device access to the operating area. SahaPilot's design goal is to close this gap: deliver autonomous material transport at roughly one-tenth of the entry cost of commercial systems, on off-the-shelf ESP32 hardware, while treating physical-layer security as a first-class design requirement rather than an add-on. This project was developed as an entry to the 15th Otomotivin Geleceği Tasarım Yarışması (OGTY 2026, theme: Industrial Mobility), and is being carried forward as an independent engineering project beyond the competition.

## 2. System Architecture

The vehicle combines two functional subsystems on shared low-cost hardware (Figure 1): a navigation subsystem consisting of a 3-channel infrared line sensor array and a forward-facing ultrasonic distance sensor, and a security subsystem consisting of a second ESP32 node that continuously scans nearby wireless traffic. Both subsystems were first validated in software before any physical construction, using Python simulations described in Section 3.

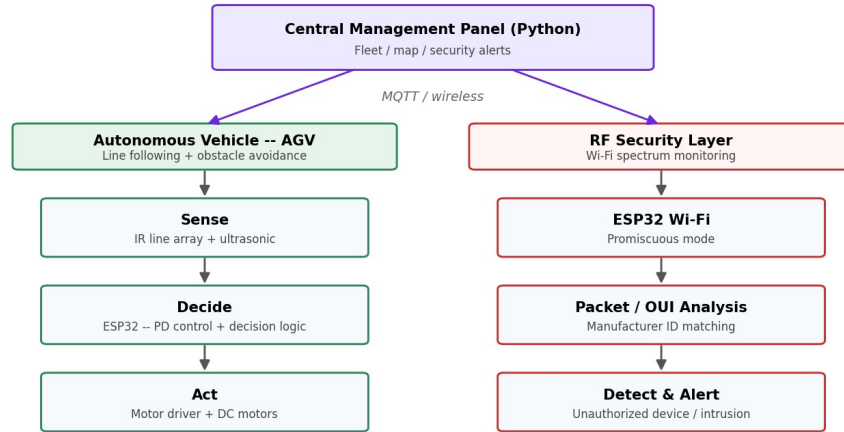


Figure 1. SahaPilot system architecture: navigation and RF security subsystems.

### 3. Methodology

#### 3.1 Lateral Control: PD versus Bang-Bang

The vehicle's lateral dynamics were modeled as a double integrator driven by a control input:

$$y'' = K_p \cdot (r - y) + K_d \cdot (r' - y')$$

where  $y$  is lateral position,  $r$  is the reference line position,  $K_p$  is the proportional gain, and  $K_d$  is the derivative (damping) gain. Three control strategies were compared under an identical initial lateral error: bang-bang control (constant-magnitude correction based only on error sign), proportional-only control, and full PD control. Bang-bang and proportional-only control both sustain persistent, undamped oscillation around the reference and never settle within the simulated window. PD control ( $K_p=4$ ,  $K_d=2$ ) converges to the reference within approximately 2 seconds of a step disturbance (Figure 3). Independently, tracking of a sinusoidal reference line under PD control closely follows the reference, while a weakly-tuned controller ( $K_p=0.5$ ,  $K_d=0.05$ ) lags significantly (Figure 2). This result directly informed the firmware design decision to implement PD control rather than threshold-based (bang-bang) steering logic.

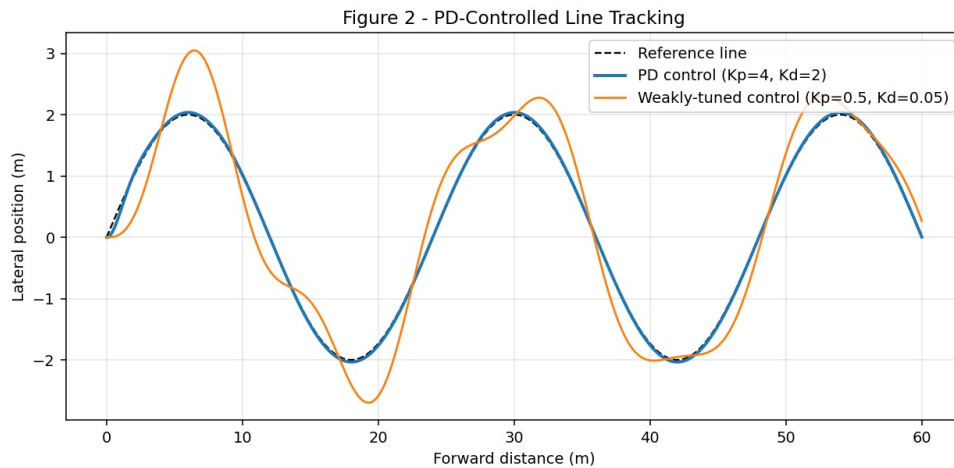


Figure 2. PD-controlled line tracking versus weakly-tuned control on a sinusoidal reference.

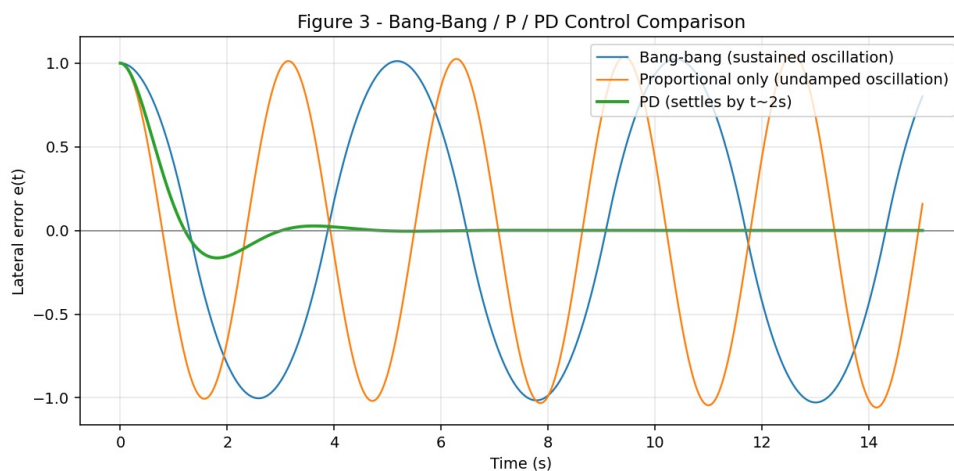


Figure 3. Step-response comparison: bang-bang, proportional, and PD control.

### 3.2 A Reference-Frame Defect in Obstacle Avoidance

An earlier implementation of the combined line-following and obstacle-avoidance logic was tested against 200 randomized scenarios rather than accepted on code inspection alone. This testing revealed 1,009 collisions across the 200 trials. Root-cause analysis showed that the forward-looking ultrasonic check and the line-centering logic operated in different frames of reference: the vehicle could correctly detect and begin avoiding an obstacle several steps ahead, only to have the line-centering behavior pull it back onto the reference line -- which frequently passed directly through the obstacle's position -- once the forward-looking check no longer registered danger from the new position. The fix adds a single explicit check prior to committing any movement: does the candidate next position collide with a known obstacle in the current row, regardless of which behavior (avoidance or line-following) proposed it. This single check reduced collisions from 1,009 to 0 across the same 200 test seeds. This defect is a useful illustration of a broader point in sensor-fusion design: two correctly-implemented subsystems can still produce unsafe joint behavior if they reason about the world in inconsistent coordinate frames.

### 3.3 RF Security Layer: OUI Matching with Consecutive-Scan Confirmation

The security subsystem identifies nearby wireless devices by the first three bytes of their MAC address (the OUI, which identifies the manufacturer) and compares this against an allow-list of known devices. A single sighting of an unrecognized OUI is not treated as an

intrusion, since real-world RF environments are noisy -- a passer-by's phone, for instance, may appear briefly and vanish. The detector instead requires the same unauthorized OUI to appear in three consecutive scans before raising an alert. This is conceptually related to the consecutive-confirmation strategies used in Constant False Alarm Rate (CFAR) radar detection to suppress spurious detections, applied here in a much simpler RF/MAC-layer context. In a 60-scan simulation in which an unauthorized device appears stochastically after scan 30 (with RSSI rising from -92 dBm to -55 dBm as it approaches), the detector confirms the intrusion within 2-3 scans of first appearance and produces zero false alarms prior to the intrusion (Figure 4).

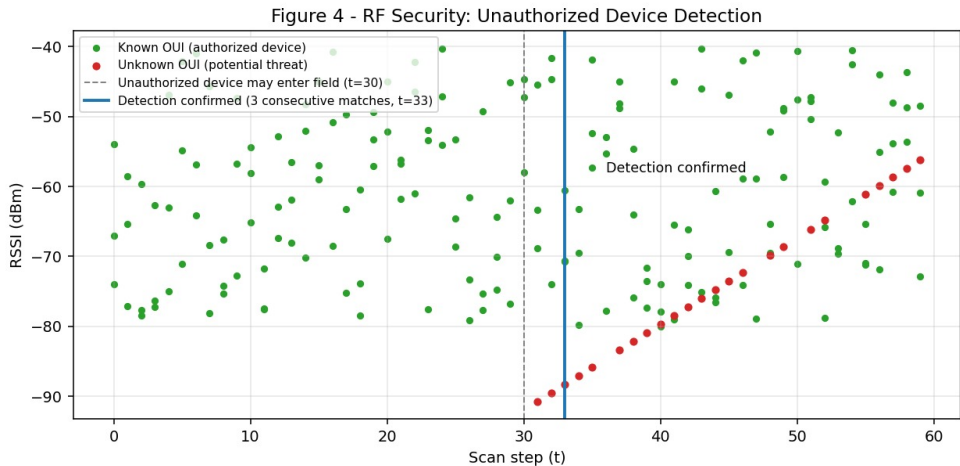


Figure 4. RF intrusion detection: RSSI of an unauthorized OUI over time, with confirmed detection point.

#### 4. Results Summary

| Metric                                                    | Result                                    |
|-----------------------------------------------------------|-------------------------------------------|
| PD vs. bang-bang settling time (step disturbance)         | ~2 s (PD) vs. no settling (bang-bang / P) |
| Navigation collisions, 200 randomized trials (before fix) | 1,009                                     |
| Navigation collisions, 200 randomized trials (after fix)  | 0                                         |
| RF intrusion detection delay                              | 2-3 scans after first appearance          |
| RF false alarms (pre-intrusion window)                    | 0                                         |
| Estimated prototype hardware cost                         | ~1,300-1,800 TRY (component-priced)       |

#### 5. Discussion and Future Work

All results reported here are simulation-based; they validate the control and detection logic prior to hardware construction, but do not yet reflect real sensor noise, motor response nonlinearity, or RF propagation effects in a real environment. The next phase of this project is a physical prototype build (ESP32, IR line sensors, ultrasonic sensor, differential drive via an L298N motor driver, and a second ESP32 node for the RF layer), with the PD gains and detection thresholds re-calibrated against real hardware. The full implementation, including firmware, simulation code, and hardware documentation, is maintained as an open-source

repository.

## 6. Conclusion

This report presented SahaPilot, a low-cost autonomous ground vehicle concept for small-scale industrial logistics, and validated its core control and security logic in simulation prior to hardware construction. A real navigation defect was found and corrected through randomized testing, and the RF security layer's consecutive-scan confirmation was shown to eliminate false positives at a small detection-latency cost. This work is being continued as an independent engineering project, with a physical prototype and field validation as the immediate next steps.

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*Code, firmware, and simulation sources: [github.com/eray2638/SahaPilot](https://github.com/eray2638/SahaPilot)*